TSA Mock

# Problem Solving

The hardest past TSA Problem Solving question of every type, from all 16 papers (2008–2023)

| Questions | Total marks | Time limit |
|---|---|---|
| 25 | 25 | 45 minutes |

## Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Every question is a real past TSA question. The year and number in brackets show where it comes from, e.g. **7 (2019 Q32)**.

## What is in this paper

At least one question of each of the 19 Problem Solving question types that have come up in past papers, chosen as the hardest of its type: 2D shapes, tiling & rotations, 3D views & cubes, Algebra & unknowns, Area, perimeter & volume, Averages, Calendars, clocks & digital displays, Charts & graphs, Choosing a method, Counting & combinations, Logic & deduction, Maps, routes & networks, Money & best value, Nets & folding, Number patterns, cycles & codes, Percentages, Rates, ratio & proportion, Reading tables & data, Speed, distance & time, Time zones & timetables.

## How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to 25 questions: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes). These are hard questions, so expect to run short of time.

**Score: _____ / 25**

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**1** (2017 Q48) Each word in a word game is scored by adding up the values of its letters. Each letter has the same value whenever it appears but different letters have different values. I know the word values for TEAR, RITE, TREE and RAT, but none of the letter values.

How many of the letter values can I now calculate?

**A** 0

**B** 1

**C** 2

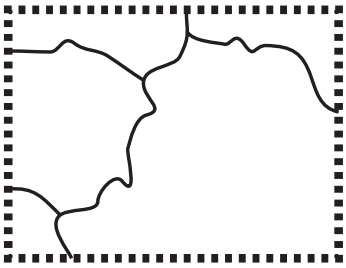
**D** 3

**E** 4

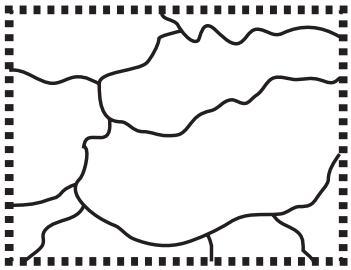
**2** (2023 Q38) The following pictures are blank maps. The continuous lines indicate country borders, the dotted lines show the borders of the maps. These maps need to be painted so that each country is filled with one colour and that no two neighbouring countries (countries with a common border) have the same colour. The fewest number of colours should be used on a given map.

Which one of the following blank maps will require 4 different colours to be painted?

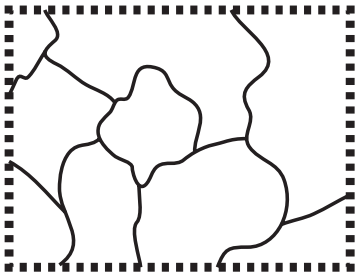
**A**



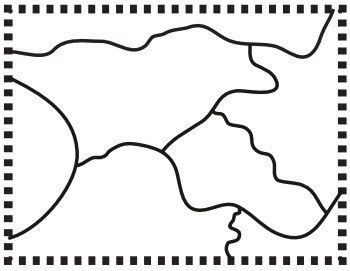
**B**



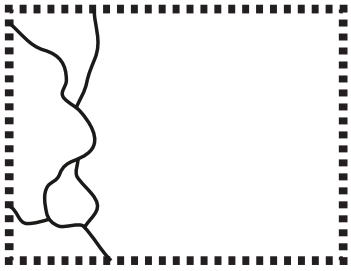
**C**



**D**

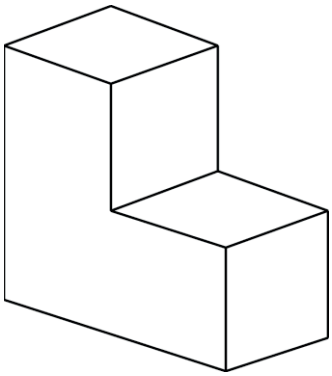


**E**



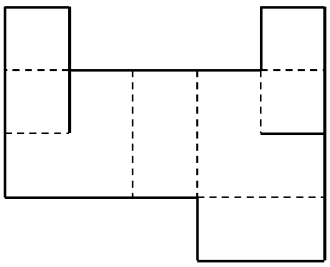
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**3** (2017 Q44) The drawing below shows an ‘L’ shaped box.

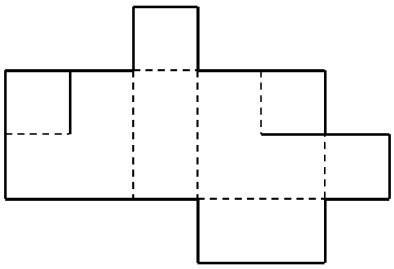


Which one of the nets below **cannot** be folded up to form the ‘L’ shaped box? (A dotted line indicates a fold and a solid line indicates a cut.)

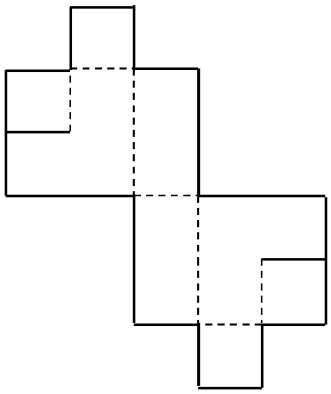
**A**



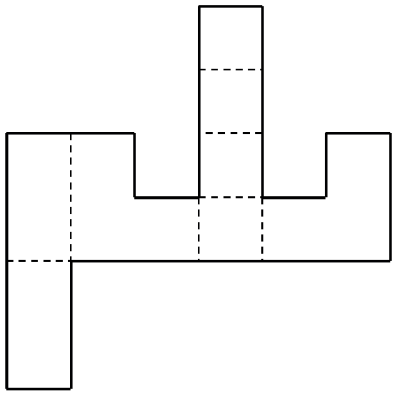
**B**



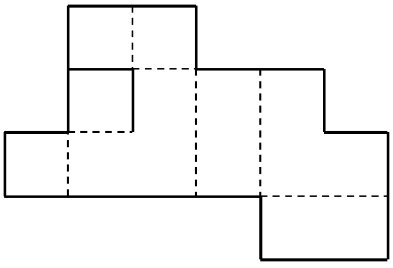
**C**



**D**

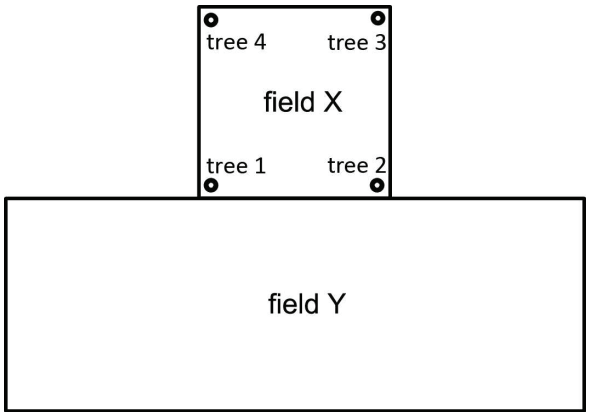


**E**



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**4** (2015 Q50) Four trees stand at the corners of square field X. They are photographed from field Y, from every possible direction, close up and from a distance, but never so that one tree stands in front of another.




In how many different orders, left to right, can the trees appear in the photographs?

**A** 1

**B** 2

**C** 3

**D** 4

**E** 5

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**5** (2014 Q50)

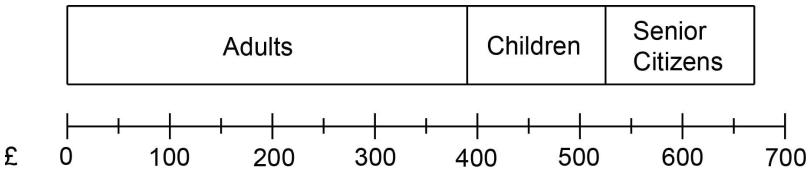
The cost of entry to the Victoria and David Museum is as follows:

Adults £7

Children £4

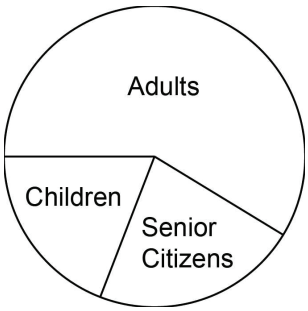
Senior Citizens £5

This chart shows the breakdown of yesterday's total takings of £679 by age group.

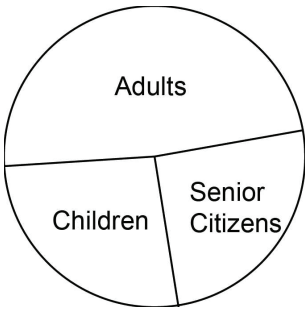



Which one of the following pie charts shows the number of visitors in each age group that paid to enter the museum yesterday?

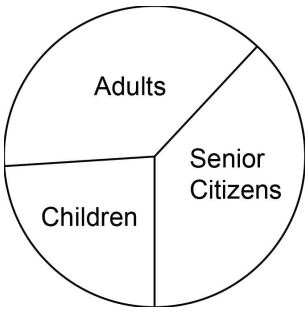
**A**



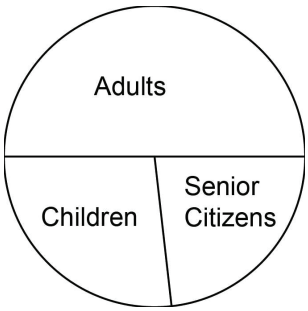

**B**



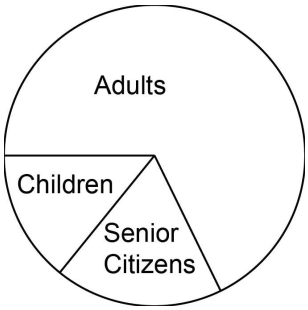

**C**




**D**

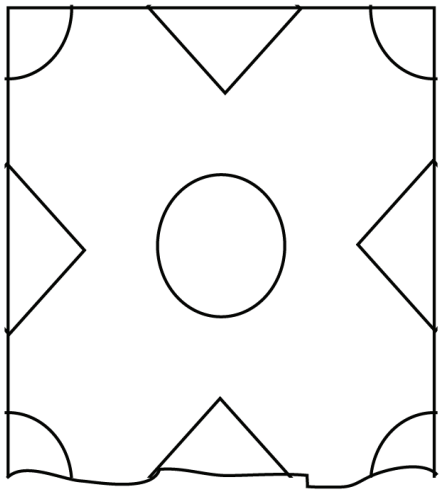



**E**




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**6** (2016 Q50) Rolls of a particular wallpaper have a repeating pattern every 60 cm of their length. Each roll is 10 m long and 50 cm wide. A wall 2.5 m high and 6.2 m wide is to be papered from ceiling to floor with single lengths of paper hung vertically (drops).



How many rolls of wallpaper will be needed so that the pattern on each drop matches the next? (No joins can be made in one vertical drop.)

**A** 3

**B** 4

**C** 5

**D** 6

**E** 7

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**7** (2020 Q43) Every Christmas the most popular children's toys become scarce. Many people turn to online auction sites to buy them instead. The table below shows the recommended retail price and the price that the listed toys have sold for online.

| *children's toy* | *recommended retail price* ($) | *online auction site* ($) |
|---|---|---|
| Moshling Tree House | 18.99 | 198.98 |
| LeapPad Explorer | 79.99 | 225.00 |
| Doggie Doo | 24.99 | 50.00 |
| Fireman Sam Pontypandy Rescue set | 29.99 | 59.99 |
| Fijit Friends | 54.99 | 99.95 |
| Monster High Lagoona's Station | 29.99 | 59.98 |
| Nerf Vortex Nitron Blaster | 44.99 | 59.99 |
| Kidizoom Twist | 49.99 | 65.95 |
| Milky the Bunny | 59.99 | 74.99 |
| Star Wars Ultimate Force Lightsaber | 39.99 | 49.95 |
| Lets Rock Elmo | 69.99 | 84.99 |
| Ninjago Fire Temple | 91.99 | 96.99 |
| Sylvanian Families | 79.95 | 192.29 |
| Vtech Innotab | 79.99 | 150.99 |
| Lego Star Wars Millennium Falcon | 132.99 | 265.98 |

How many of the toys above have increased in price by more than 100%?

**A** 3

**B** 4

**C** 5

**D** 6

**E** 7

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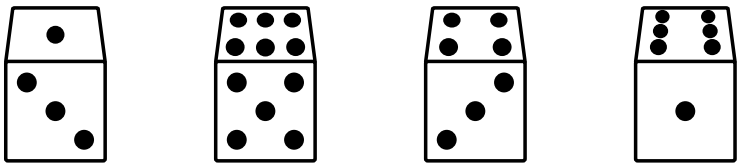
**8** (2010 Q49) All the telephone numbers in my village have six digits and begin with 58. I have accidentally torn up and thrown away a piece of paper on which I had written the telephone number of a friend who lives nearby. The only piece I can find is the scrap shown below:




How many different phone numbers in my village could be consistent with this extract?

**A** 2
**B** 10
**C** 11
**D** 12
**E** 22

**9** (2010 Q50) On a normal die, opposing faces add up to 7. Four views of the same die on which the numbers are arranged differently from a normal die are shown below. The unseen face has the number two on it.

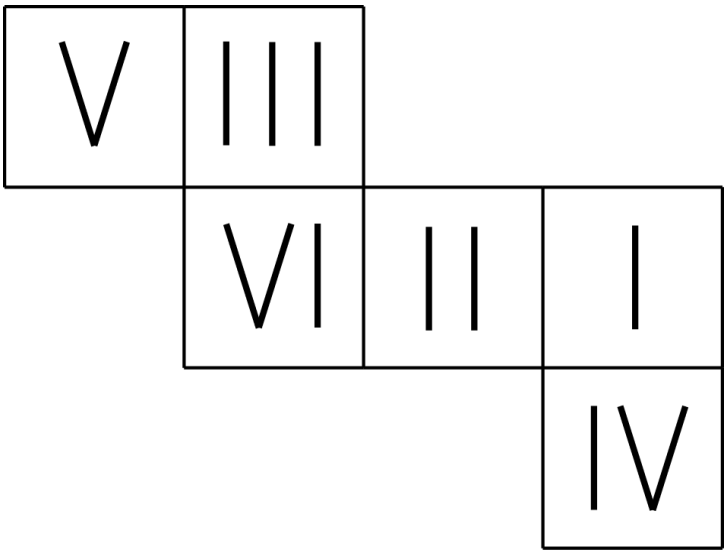


How many spots are on the face opposite the two on this die?

**A** 1
**B** 3
**C** 4
**D** 5
**E** 6

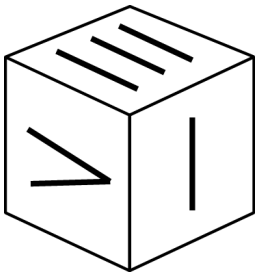
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**10** The net below can be folded to make a Roman numerical dice.
(2019 Q50)

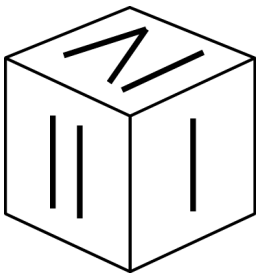



Which one of the following dice could have been made as a result of folding the net?

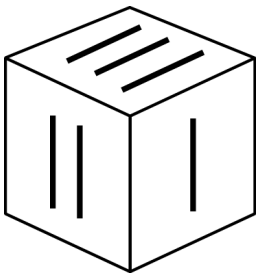
**A**



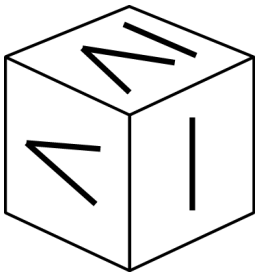

**B**



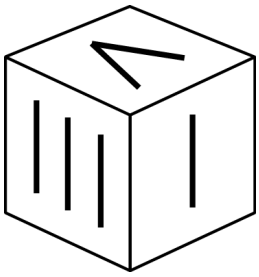

**C**




**D**

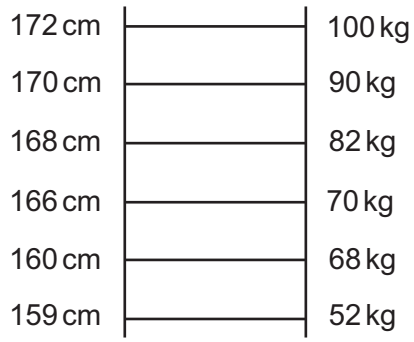



**E**

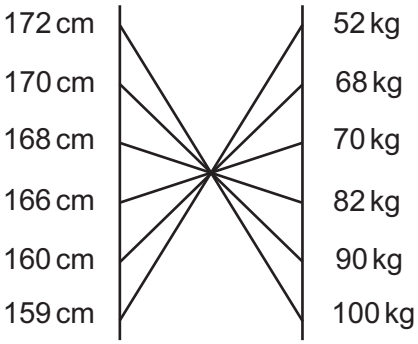



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**11** (2018 Q50) Keith likes to draw ‘bump charts’ to look at relationships. For instance, if 6 individuals were ordered by height, and it happened that the order was the same by weight, the bump chart would look like this:

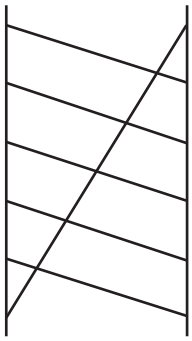



Or if height was maintained for order on the left hand but lightness rather than heaviness on the right hand, the bump chart would be this:

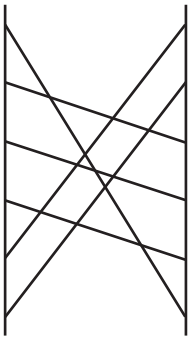



If one axis is reversed in order (as in heaviness goes to lightness), which one of the following bump charts would **not** be symmetrical about a vertical line midway between the axes?

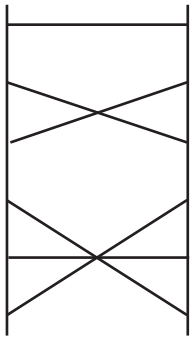
**A**



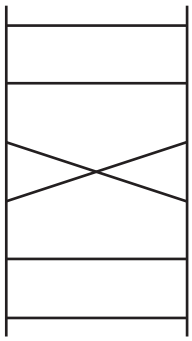
**B**



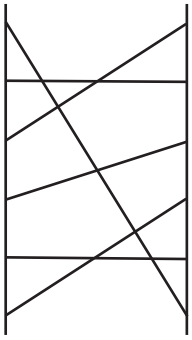
**C**



**D**

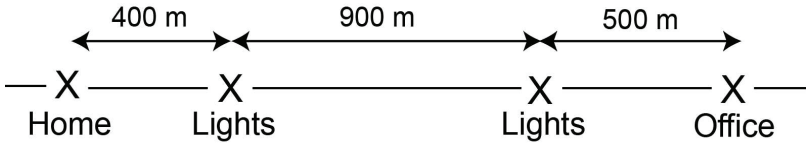


**E**



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**12** (2014 Q6) Central Avenue is a long straight road, several kilometres long. Both my home and my office are on Central Avenue, and the diagram below shows their positions, together with two sets of traffic lights that lie in between.




Both sets of lights turn to green for Central Avenue traffic simultaneously, showing green for exactly two minutes, turning green again exactly two minutes later.

I cycle to work every morning at a steady speed of 5 metres per second (except, of course, whilst waiting at traffic lights).

What is the **longest** journey time I can expect the journey to work to take?

**A** 6 minutes

**B** 7 minutes

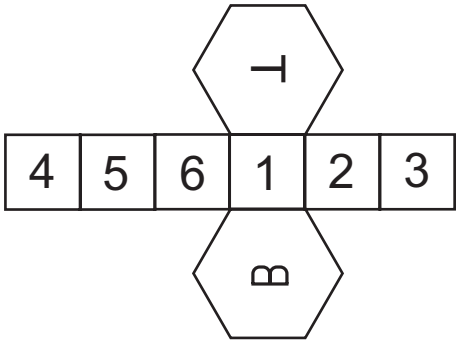
**C** 8 minutes

**D** 9 minutes

**E** 10 minutes

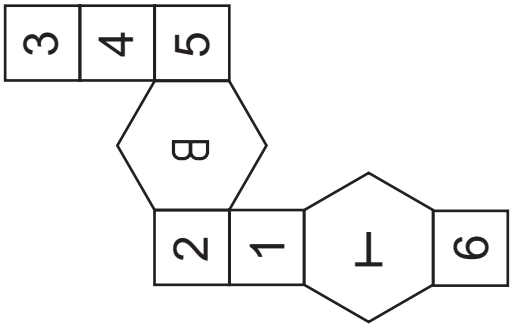
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**13** Consider the following net, which forms a hexagonal prism:
(2019 Q44)

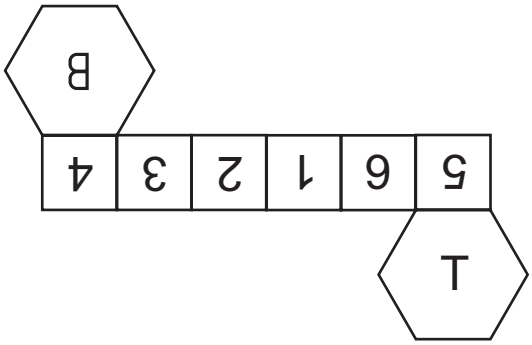



Which one of the following nets will form an identical prism?

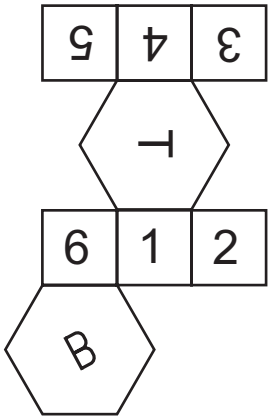
**A**



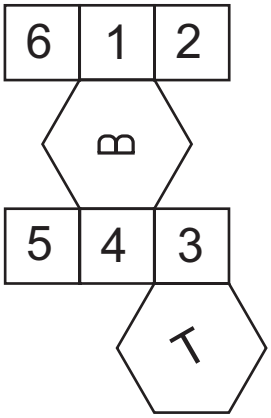

**B**



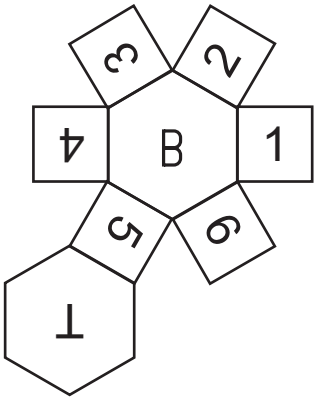

**C**




**D**




**E**




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**14** (2018 Q25) The President of Bugaria is elected using the supplementary vote system, which is conducted as follows:

Voters vote for their first-choice and second-choice candidates (though a second choice is optional).

The two candidates with the most first-choice votes go through to a second stage, whilst the others are eliminated. The second-choice votes for these two from the ballot papers of the eliminated candidates are added to their first-choice votes, and the candidate with the greater total is elected President.

The table below shows a summary of the votes cast for the five candidates in last week's Bugarian Presidential Election. For example, 37 862 voters cast their first vote for Sam and their second vote for Val.

| | *first-choice votes* | | | | |
|---|---|---|---|---|---|
| *second-choice votes* | **Sam** | **Val** | **Kim** | **Pat** | **Jo** |
| Sam | – | 37 304 | 34 493 | 38 206 | 22 871 |
| Val | 37 862 | – | 25 517 | 27 818 | 20 143 |
| Kim | 23 148 | 31 690 | – | 30 247 | 42 620 |
| Pat | 26 493 | 24 485 | 29 681 | – | 31 354 |
| Jo | 28 506 | 27 530 | 25 308 | 24 763 | – |
| no second choice | 822 | 1027 | 1263 | 952 | 848 |

Who has been elected President of Bugaria?

**A** Sam

**B** Val

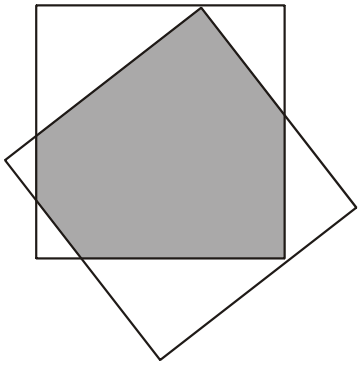
**C** Kim

**D** Pat

**E** Jo

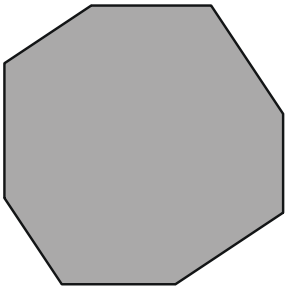
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**15** (2011 Q25) Various shapes may be made by overlapping two transparent squares, for example the dark shaded hexagon shown below:

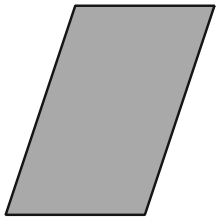


Which one of the following shapes cannot be made as the overlap of two transparent squares?

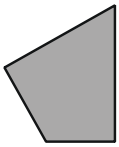
**A**



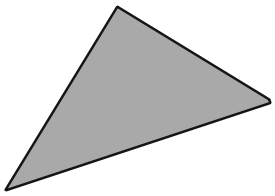
**B**



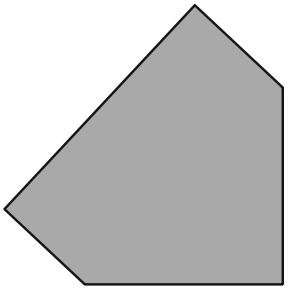
**C**



**D**



**E**



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**16** (2021 Q50) The company that I currently work for pays its employees monthly. Pay day is the fourth Thursday of every month or the 26th, whichever is earlier. Since I joined the company the dates on which I have been paid (in order) have been:

$24^{th}$

$24^{th}$

$26^{th}$

$26^{th}$

$23^{rd}$

$26^{th}$

$25^{th}$ (last month)

On what date will I be paid this month?

**A** $22^{nd}$

**B** $23^{rd}$

**C** $24^{th}$

**D** $25^{th}$

**E** $26^{th}$

**17** (2013 Q50) The housing estate on which Andrew lives has streets of variable length. In each street the houses are directly opposite each other, with the same number of houses on each side. However, the numbering of the houses is inconsistent.

Some streets are numbered consecutively on each side, starting at one end of the street, going up to the end and then back down the other side.

Other streets have the odd numbers on one side, starting from 1, and the evens on the other side, starting with 2 at the same end.

Andrew's friend Amy lives on the same estate at 25 Acacia Avenue, which is not the end house in the street.

If Andrew knows the number of the house opposite Amy's, which one of the following statements is definitely true?

**A** He knows the exact number of houses in the street.

**B** He knows that the number of houses in the street is one of two numbers.

**C** If the house opposite Amy's is 26, he knows the number of houses in the street.

**D** If he knows the number of the house next door to Amy, he can work out the number of houses in the street.

**E** He knows the numbers of the houses immediately adjoining Amy's.

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**18** (2023 Q36) Becky works from home making soft toys which she sells online. Each one takes her exactly 16 minutes to make and she takes a 5-minute break between finishing one item and starting the next. She always takes a 40-minute break for her lunch. Today she is starting on a new batch of toys and is planning to work from 08:30 until 17:30.

How many soft toys will she have completed by 17:30?

**A** 23

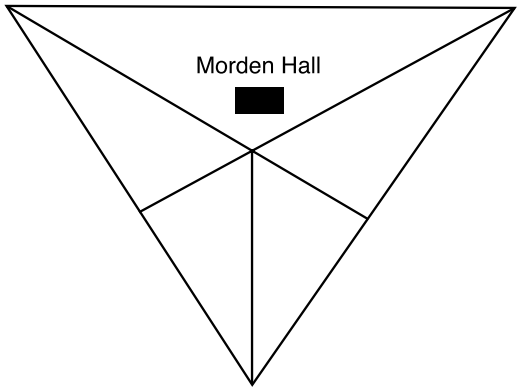
**B** 24

**C** 25

**D** 26

**E** 27

**19** (2011 Q44) The diagram below shows Morden Hall and the network of paths within the grounds.



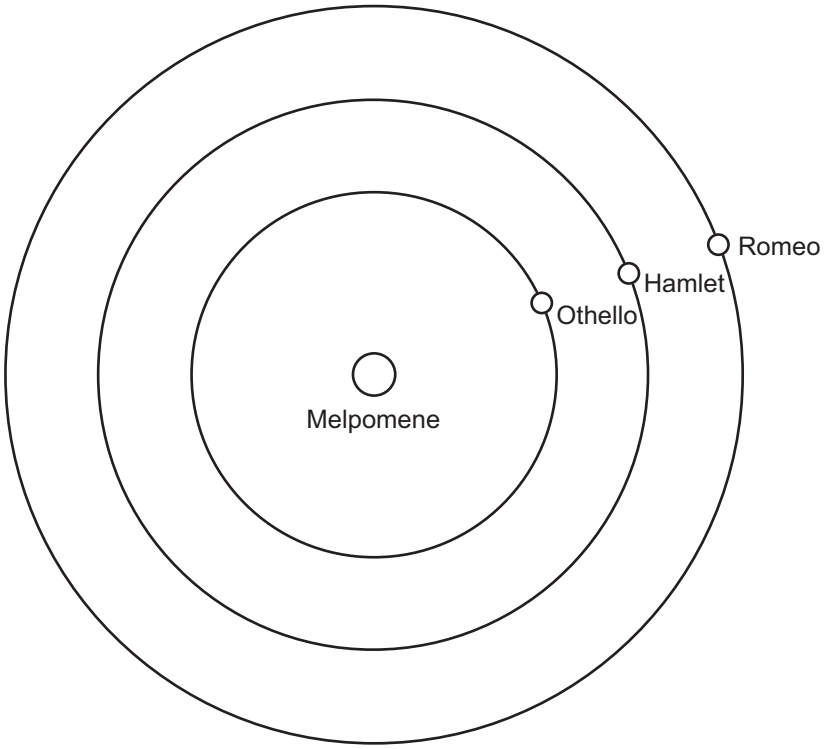

When the owner takes his dog for a walk he always starts and finishes at the intersection outside the front door and (keeping to the paths) walks one complete triangular route.

How many different routes does he have to choose from? (Clockwise and anticlockwise around the same triangle counts as one route only.)

**A** 5
**B** 7
**C** 10
**D** 11
**E** 12

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**20** (2016 Q48) The planet Melpomene has three moons, Othello, Hamlet and Romeo, each moving at a constant speed in the same direction in circular orbits.




Othello takes 20 days to complete one orbit of Melpomene, whereas Hamlet takes 45 days and Romeo takes 120 days.

How often are Melpomene and all three of its moons in line with each other in the order shown in the diagram?

**A** every 36 days

**B** every 72 days

**C** every 132 days

**D** every 180 days

**E** every 216 days

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**21** (2023 Q31) Imani is looking to book a four-night stay in May at the Highgate Hotel, to include a Sunday night so that she can attend an evening orchestral concert.

The table below shows the daily rates for the Highgate Hotel.

**Highgate Hotel**
Daily rates for a single room with breakfast
**May**

| *Monday* | *Tuesday* | *Wednesday* | *Thursday* | *Friday* | *Saturday* | *Sunday* |
|---|---|---|---|---|---|---|
| 1st **£30** | 2nd **£40** | 3rd **£50** | 4th **£50** | 5th **£20** | 6th **£50** | 7th **£40** |
| 8th **£30** | 9th **£40** | 10th **£50** | 11th **£50** | 12th **£20** | 13th **£50** | 14th **£40** |
| 15th **£30** | 16th **£40** | 17th **£60** | 18th **£40** | 19th **£30** | 20th **£40** | 21st **£40** |
| 22nd **£30** | 23rd **£20** | 24th **£50** | 25th **£40** | 26th **£30** | 27th **£60** | 28th **£30** |
| 29th **£40** | 30th **£50** | 31st **£50** | | | | |

**May Special Offers**
Book Saturday and Sunday and stay Friday for free.
Book Sunday and Monday and stay Tuesday for free.
**Only one special offer can be used per booking.**

What is the least amount Imani needs to pay for her four-night stay?

**A** £100

**B** £110

**C** £120

**D** £130

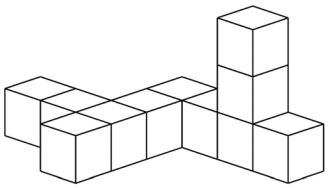
**E** £140

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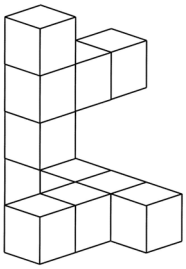
**22** (2012 Q44) Five pictures of solids made by sticking cubes together are shown below.

Which one of the following (A-E) is a different shape from **all** the others?

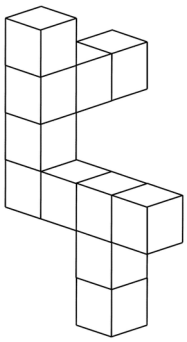
**A**



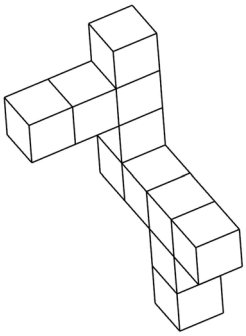
**B**



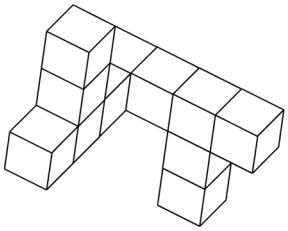
**C**



**D**



**E**



**23** (2020 Q46) My dog's coat had been looking dull, so I bought some conditioning tablets. The pet shop stocks these in boxes of 300 tablets, and for a dog the size of mine the recommendation (which I have followed) is 6 tablets per day.

Now, several months later, his coat is much shinier. Opening a new box of tablets today, I have decided to reduce his dosage. I shall just give him 6 tablets per day, twice a week, and on the remaining days each week he can have 4 tablets each day.

What is the maximum number of extra days that this box of conditioning tablets will now last?

**A** 10 days

**B** 15 days

**C** 16 days

**D** 24 days

**E** 25 days

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**24** (2013 Q36) Last month, in an attempt to improve my fitness, I began to run daily. The maximum distance I have time to fit in on any particular day is 8 miles, but my aim is to maintain an average of at least 5 miles per day.

After 15 days my average was just under 6 miles per day, then the following day I increased my average to just over 6. Since then (for the last 9 days, up to and including today) I have been able to run only 2 miles each day.

What is the **minimum** number of days from now until I can bring my average back up to 5 miles per day?

**A** 2 days

**B** 4 days

**C** 9 days

**D** 18 days

**E** 26 days

**25** (2011 Q38) John wishes to have 500 copies of a brochure printed and is offered a choice of two price structures:

1. An equal cost for each brochure printed, regardless of number.
2. An initial fixed cost followed by a lower cost than structure 1 for each brochure.

Which one of the following sets of information would be sufficient for him to be able to decide which of the two price structures would be least expensive?

**A** The number of brochures for which the total bill would be the same by both structure 1 and structure 2.

**B** An example showing the cost of 500 brochures for structure 1 and 200 brochures using structure 2.

**C** The cost per brochure using structure 1 and the initial cost using structure 2.

**D** An example showing the cost of 200 brochures using structure 1 and 200 brochures using structure 2.

**E** The cost of 1 brochure using structure 1 and 2 brochures using structure 2.

**END OF TEST**

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